

Sequences

Test · 26 marks total

Name: _____

Date: _____

Score: _____ / 26

Section A — Term-to-term rules

A1. Write the next term of 12, 17, 22, 27, ...

[1 mark]

A2. Write the next term of 60, 52, 44, ...

[1 mark]

A3. State the term-to-term rule for 3, 6, 12, 24, ...

[1 mark]

A4. State the term-to-term rule for 90, 81, 72, 63, ...

[1 mark]

Section B — Special sequences

B1. Write the first five square numbers.

[2 marks]

B2. Write the first five triangular numbers.

[2 marks]

B3. This is a Fibonacci-type sequence: 3, 5, 8, 13, ... Find the next two terms.

[2 marks]

Section C — Finding the n th term

C1. Find the n th term of the sequence 2, 5, 8, 11, ...

[3 marks]

C2. Find the n th term of the sequence 40, 33, 26, 19, ...

[3 marks]

Section D — Using the n th term

D1. The n th term of a sequence is $4n + 5$. Find the 12th term.

[2 marks]

D2. The n th term of a sequence is $6n - 4$. Is 100 a term in this sequence? Show your working.

[3 marks]

D3. The first term of a sequence is 9 and the term-to-term rule is "add 5". Find the n th term, and use it to find the 20th term.

[5 marks]